

Multiple Choice (10 marks)

1. Miranda enlarged a picture proportionally. Her original picture is 4 cm wide and 6 cm long. If the new, larger picture is 10 cm wide, what is its length?
 - (a) 22 cm
 - (b) 15 cm
 - (c) 16 cm
 - (d) 10 cm
 - (e) 12 cm
2. For how many positive integers k does the ordinary decimal representation of the integer $k!$ end in exactly 99 zeros?
 - (a) Twenty
 - (b) Three
 - (c) None
 - (d) Four
 - (e) Five
3. Compute covariance of $x=(1,2,3,4)$, $y=(2,3,4,5)$
 - (a) 0.50
 - (b) 2.00
 - (c) 3.00
 - (d) 0.75
 - (e) 1.25
4. Find the sum of $\sum_{n=1}^{\infty}(\cos(1/n^2) - \cos(1/(n+1)^2))$
 - (a) -2.459
 - (b) -1.459
 - (c) -0.159
 - (d) 2.459
 - (e) 0.959
5. Each time Rami turned the dial on a machine, the dial moved 1 degree. Rami turned the dial 10 times. What is the total number of degrees the dial moved?
 - (a) 270

- (b) 100
- (c) 90
- (d) 20
- (e) 110

True / False (10 marks)

Answer True or False

6. True or False: The number of ways to arrange 6 pairs of parentheses such that they are balanced is 42.
7. True or False: The value of the series $\sum_{n=0}^{\infty} (-1)^n \frac{1}{3n+1}$ is approximately 1.1356488482647211.
8. True or False: If each puzzle cost 6, *the total cost of 934 puzzles sold would be 5,764.*
9. True or False: The difference in elevation between Talon Bluff and Salt Flats is 695 feet.
10. True or False: $(1+i)^{10} = 32i$, demonstrating the application of De Moivre's theorem in complex number exponentiation.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Compute

$$\sum_{k=2}^{63} \log_2 \left(1 + \frac{1}{k} \right) \log_k 2 \log_{k+1} 2.$$

12. Discuss the various arrangements possible for a panel consisting of three students and six faculty members seated at a table. Provide a detailed analysis of the permutations involved.

End of Paper